

Table 1: Method comparison on CORE4D-G1 (all 50 paper sequences), mean over cases. **SPIDER-CEM** (ours) vs **OmniRetarget** (our upstream kinematic input), **SBTO**, and **GMR**. Physical contact/penetration are frame fractions from the same MuJoCo contact detector; max penetration is the deepest hand-object interpenetration. $\uparrow/\downarrow$: higher/lower is better; best per row in bold, second best underlined. OmniRetarget (kinematic, no collision) penetrates heavily (44.6% pen@5mm, 21.6 mm max) at modest contact; SPIDER-CEM’s physics optimisation raises contact to 72% while cutting pen to 7.6%. GMR’s low pen@5mm is instead a ‘barely-touches’ artifact (contact@5mm $\approx$ 0.24%); its max penetration (23.5 mm) still exceeds ours.

Metric	SPIDER-CEM	OmniRetarget	SBTO	GMR
Phys. contact@5mm (%) $\uparrow$	72.47	21.86	<u>34.37</u>	0.22
Phys. penetration@5mm (%) $\downarrow$	<u>7.59</u>	44.63	36.59	4.65
Max phys penetration (mm) $\downarrow$	12.01	<u>21.61</u>	25.36	23.51
Obj speed max (m/s) $\downarrow$	1.29	2.77	<u>1.76</u>	1.83
Foot slip max (m) $\downarrow$	0.87	1.33	1.20	<u>0.97</u>